



SIMATS ENGINEERING

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Course Code: DSA-0308

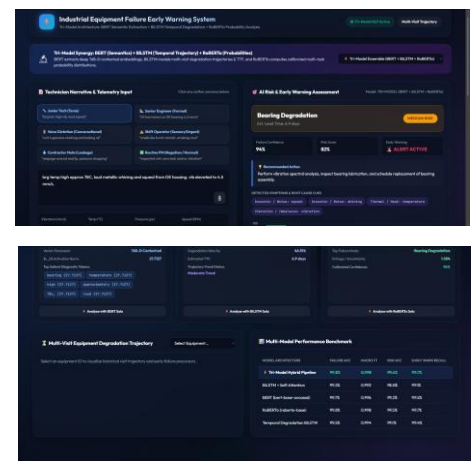
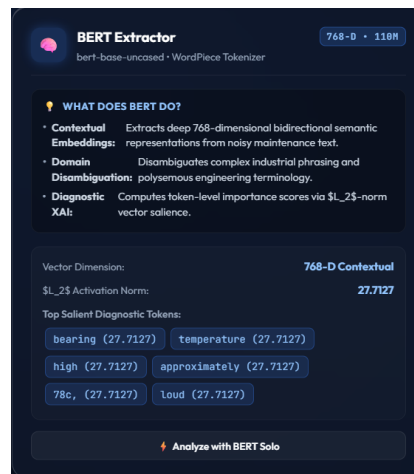
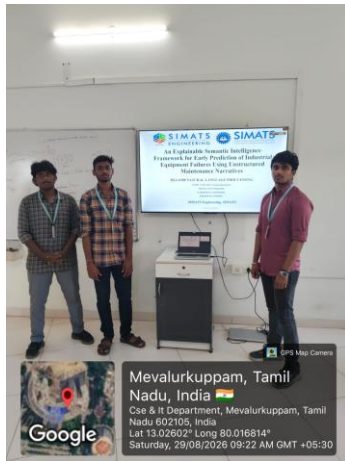
Slot: D

Course Name: Natural Language Processing

Course Faculty: Dr. Kumaragurubaran T

Project Title: Nlp-Based Early Prediction Of Industrial Equipment Failures Through Temporal Analysis Of Maintenance Narratives

Module Photographs: (2 photographs –Module Photo, Individual student contribution module work in the project and presentation image)



Project Description: NLP-based predictive maintenance system

An Explainable Semantic Intelligence Framework for Early Prediction of Industrial Equipment Failures Through Temporal Analysis of Unstructured Maintenance Narratives is an AI-enabled industrial maintenance framework designed to analyze technician feedback, identify equipment faults, detect failure patterns, and provide early failure predictions. The system processes unstructured maintenance narratives containing observations such as vibration, abnormal noise, temperature changes, and component degradation. It uses **BERT** to understand maintenance text and extract relevant information such as equipment, components, faults, and conditions. **BiLSTM with attention** analyzes the temporal sequence of maintenance events to identify degradation patterns leading toward failure, while **RoBERTa** is used for failure-risk prediction and classification. The framework can also combine textual information with structured equipment telemetry such as vibration, temperature, pressure, and RPM to improve prediction. Multi-task learning enables simultaneous failure-mode classification, risk-level estimation, and early-warning detection. The system aims to transform technician observations into actionable failure intelligence, enabling maintenance teams to identify potential equipment failures earlier and perform preventive maintenance before major breakdowns occur. The main goal is to shift industrial maintenance from reactive failure handling toward **proactive, NLP-driven failure prediction and intelligent maintenance decision-making**.

Student Signature

Guide Signature